Lesson 09 Demo 04

Configuring Terraform Remote State Backend

Objective: To configure and manage Terraform state using the remote enhanced backend with Terraform Cloud for ensuring efficient and collaborative state management

Tools required: Visual Studio Code

Prerequisites: Terraform Cloud account
Ensure you have created and implemented the AWS access key and secret key before starting this demo. Refer to Lesson 08 Assisted Practice 02 for detailed steps.

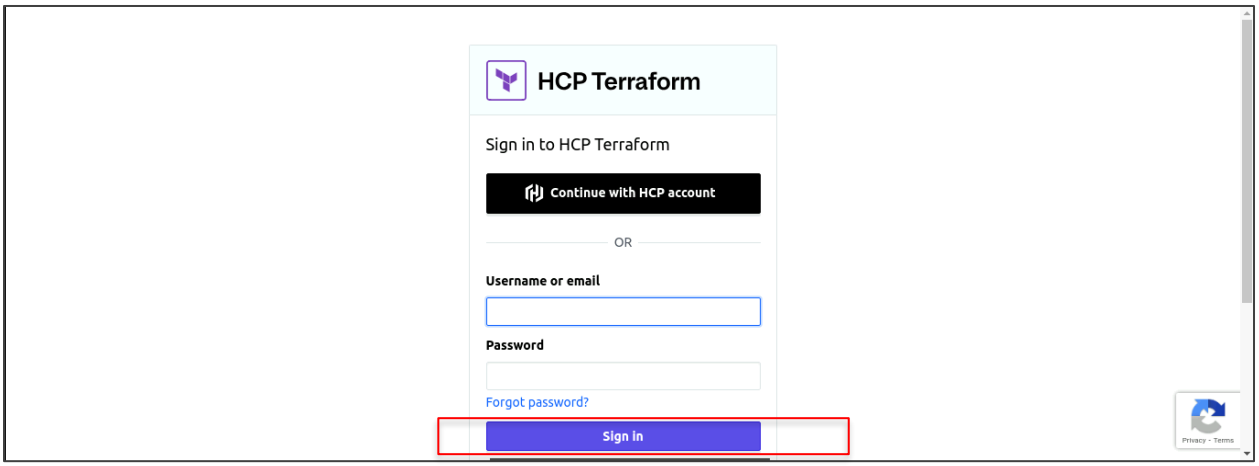
Note: The folder structure created in the previous demos is used here. It is also included in the resources section of LMS. Please refer to Lesson 10 Demo 01.

Steps to be followed:

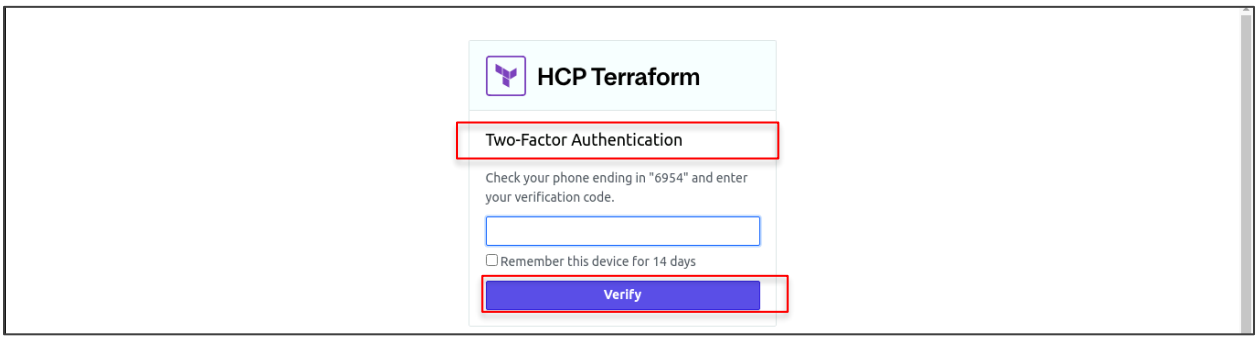
1. Sign in to Terraform Cloud platform
2. Update the Terraform configuration to use remote enhanced backend
3. Re-initialize Terraform and validate the remote backend with Terraform Cloud
4. Provide secure credentials for remote runs
5. Remove existing resources with the terraform destroy command

Step 1: Sign in to Terraform Cloud platform

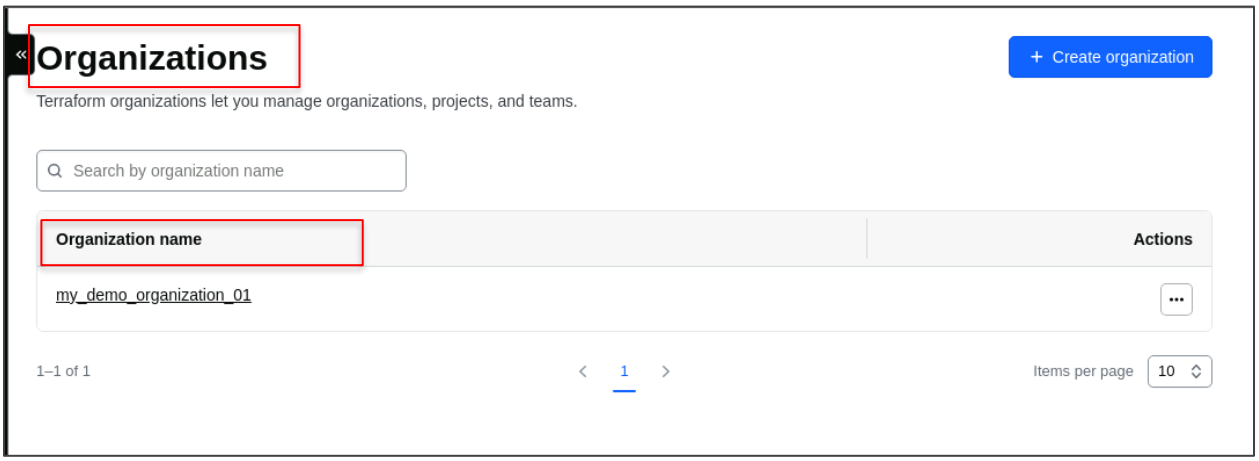
1.1 Enter the required details and click on **Sign In** by using the following URL:
https://app.terraform.io/session



1.2 Verify the **Two-Factor Authentication**



1.3 The **Organizations** menu will automatically open. Take note of the **Organization name**.



Step 2: Update the Terraform configuration to use remote enhanced backend

2.1 Edit your **terraform.tf** file to include the remote backend configuration by using the following code:

```
terraform {
  backend "remote" {
    hostname     = "app.terraform.io"
     # Replace " my_demo_organization_01" with your actual Terraform Cloud
     organization name
    organization = "my_demo_organization_01"
    workspaces {
      name = "my-aws-app"
```

```
    }
  }
}
```

```
main.tf    terraform.tf ×    variables.tf    terraform.tfstate
terraform.tf
terraform {
  backend "s3" {
    bucket = "myterraformstatedemo"
    key    = "prod/aws_infra"
    region = "us-east-1"

    # Replace this with your DynamoDB table name!
    dynamodb_table = "terraform-locks"
    encrypt        = true
  }
```

```
terraform.tf
terraform {
  backend "remote" {
    hostname     = "app.terraform.io"
    organization = "my_demo_organization_01"
    workspaces {
      name = "my-aws-app"
    }
  }
```

2.2 Format the Terraform configuration file by using the following command:
terraform fmt

```
sakshiguptasimp@ip-172-31-22-2:~/Desktop/Terraform$ terraform fmt
terraform.tf
sakshiguptasimp@ip-172-31-22-2:~/Desktop/Terraform$
```

Step 3: Re-initialize Terraform and validate the remote backend with Terraform Cloud

3.1 Re-initialize the backend using the following command:

terraform init -reconfigure

```
sakshiguptasimp@ip-172-31-22-2:~/Desktop/Terraform$ terraform init -reconfigure
Initializing modules...

Initializing the backend...

Successfully configured the backend "remote"! Terraform will automatically
use this backend unless the backend configuration changes.

Initializing provider plugins...
```

```
PROBLEMS    OUTPUT    TERMINAL    DEBUG CONSOLE    PORTS

- Reusing previous version of hashicorp/tls from the dependency lock file
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/random v3.1.0
- Using previously-installed hashicorp/local v2.1.0
- Using previously-installed hashicorp/tls v3.1.0
- Using previously-installed hashicorp/aws v3.76.1
- Using previously-installed hashicorp/http v2.1.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
sakshiguptasimp@ip-172-31-22-2:~/Desktop/Terraform$
```

3.2 Apply the configuration to the remote backend using the following command:

terraform apply

```
sakshiguptasimp@ip-172-31-22-2:~/Desktop/Terraform$ terraform apply
Running apply in the remote backend. Output will stream here. Pressing Ctrl-C
will cancel the remote apply if it's still pending. If the apply started it
will stop streaming the logs, but will not stop the apply running remotely.

Preparing the remote apply...

To view this run in a browser, visit:
https://app.terraform.io/app/my_demo_organization_01/my-aws-app/runs/run-jGiC1V41yFLKVkxW

Waiting for the plan to start...

Terraform v1.1.6
on linux_amd64
Initializing plugins and modules...
```

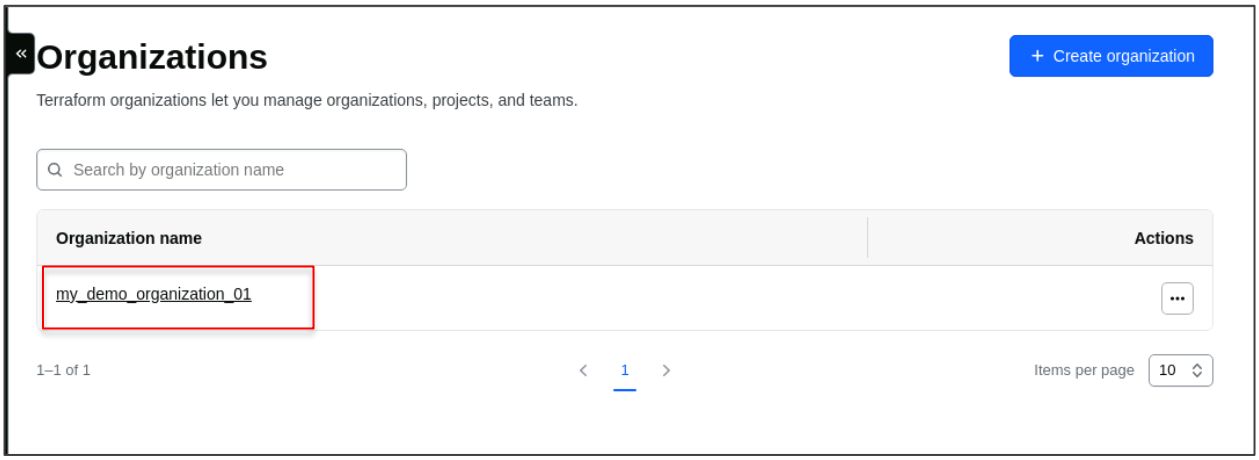
3.3 When prompted, approve the changes by typing **yes**

```
PROBLEMS    OUTPUT    TERMINAL    DEBUG CONSOLE    PORTS

          + volume_size           = (known after apply)
          + volume_type           = (known after apply)
        }
    }

Plan: 30 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + public_dns                 = (known after apply)
  + public_dns_server_subnet_1 = (known after apply)
  + public_ip                  = (known after apply)
  + public_ip_server_subnet_1  = (known after apply)
  + size                       = "t2.micro"

Do you want to perform these actions in workspace "my-aws-app"?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes
```

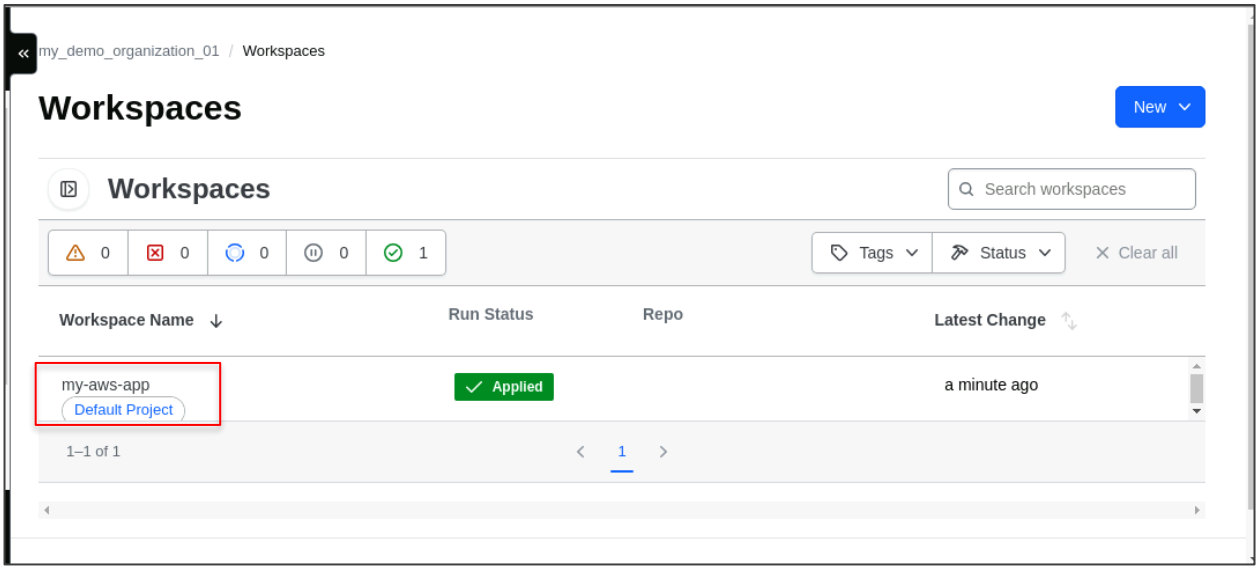
```
PROBLEMS    OUTPUT    TERMINAL    DEBUG CONSOLE    PORTS                    terraform

aws_subnet.private_subnets["private_subnet_1"]: Creating...
aws_subnet.private_subnets["private_subnet_2"]: Creating...
aws_subnet.public_subnets["public_subnet_1"]: Creating...
aws_subnet.public_subnets["public_subnet_3"]: Creating...
aws_security_group.ingress-ssh: Creating...
aws_security_group.vpc-ping: Creating...
aws_security_group.vpc-web: Creating...
aws_internet_gateway.internet_gateway: Creation complete after 0s [id=igw-0935512912e66c367]
aws_eip.nat_gateway_eip: Creating...
aws_subnet.private_subnets["private_subnet_3"]: Creation complete after 0s [id=subnet-0285e954d5276877b]
aws_route_table.public_route_table: Creating...
aws_subnet.private_subnets["private_subnet_2"]: Creation complete after 0s [id=subnet-0c9cce476fd244ce5]
aws_subnet.private_subnets["private_subnet_1"]: Creation complete after 0s [id=subnet-01567f09b7d17aeb7]
aws_eip.nat_gateway_eip: Creation complete after 1s [id=eipalloc-000a281852ff09f80]
aws_route_table.public_route_table: Creation complete after 1s [id=rtb-0c28c5ef816667ff7]
aws_security_group.vpc-ping: Creation complete after 2s [id=sg-04d7b6e8a8df09643]
aws_security_group.vpc-web: Creation complete after 2s [id=sg-0ba47019920676600]
aws_security_group.ingress-ssh: Creation complete after 2s [id=sg-0b8dc2374b6814fea]
```

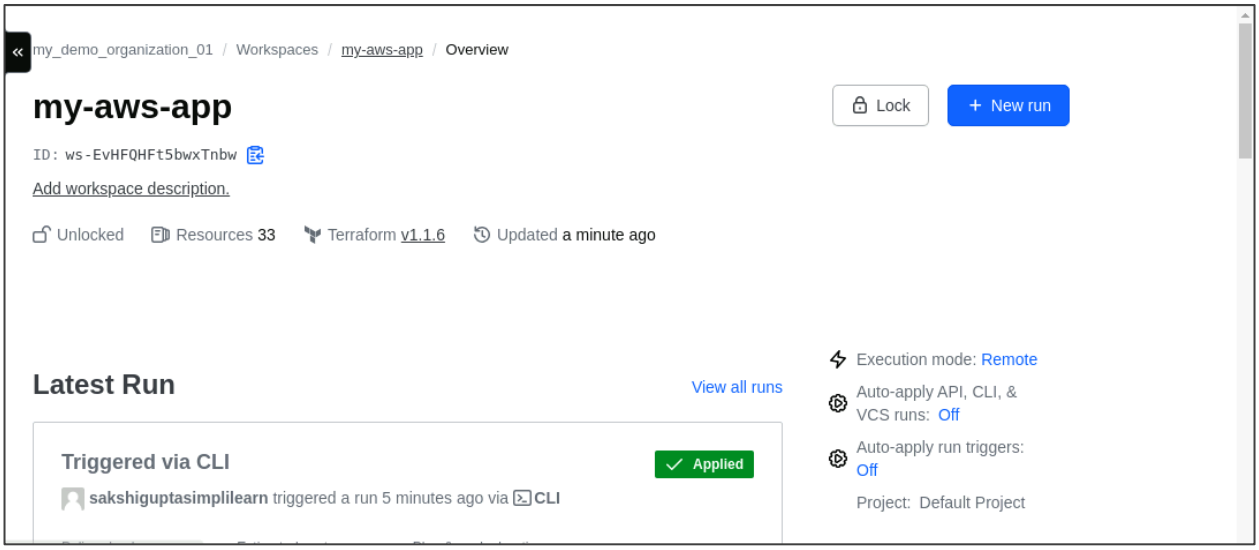
```
PROBLEMS   OUTPUT   TERMINAL   DEBUG CONSOLE   PORTS                                   bash

aws_route_table.private_route_table: Creating...
aws_route_table.private_route_table: Creation complete after 1s [id=rtb-06adcf5d46a0a60de]
aws_route_table_association.private["private_subnet_1"]: Creating...
aws_route_table_association.private["private_subnet_2"]: Creating...
aws_route_table_association.private["private_subnet_3"]: Creating...
aws_route_table_association.private["private_subnet_2"]: Creation complete after 0s [id=rtbassoc-01989bdfb41e77f2c]
aws_route_table_association.private["private_subnet_3"]: Creation complete after 1s [id=rtbassoc-0f9b7f543dfcfa244]
aws_route_table_association.private["private_subnet_1"]: Creation complete after 1s [id=rtbassoc-00d28a447954f964f]

Apply complete! Resources: 30 added, 0 changed, 0 destroyed.

Outputs:

public_dns = "ec2-44-220-82-222.compute-1.amazonaws.com"
public_dns_server_subnet_1 = "ec2-3-95-172-42.compute-1.amazonaws.com"
public_ip = "44.220.82.222"
public_ip_server_subnet_1 = "3.95.172.42"
size = "t2.micro"
sakshiguptasimp@ip-172-31-22-2:~/Desktop/Terraform$
```

3.4 Go to Terraform Cloud to validate the remote backend. Click on the organization in which you are working.

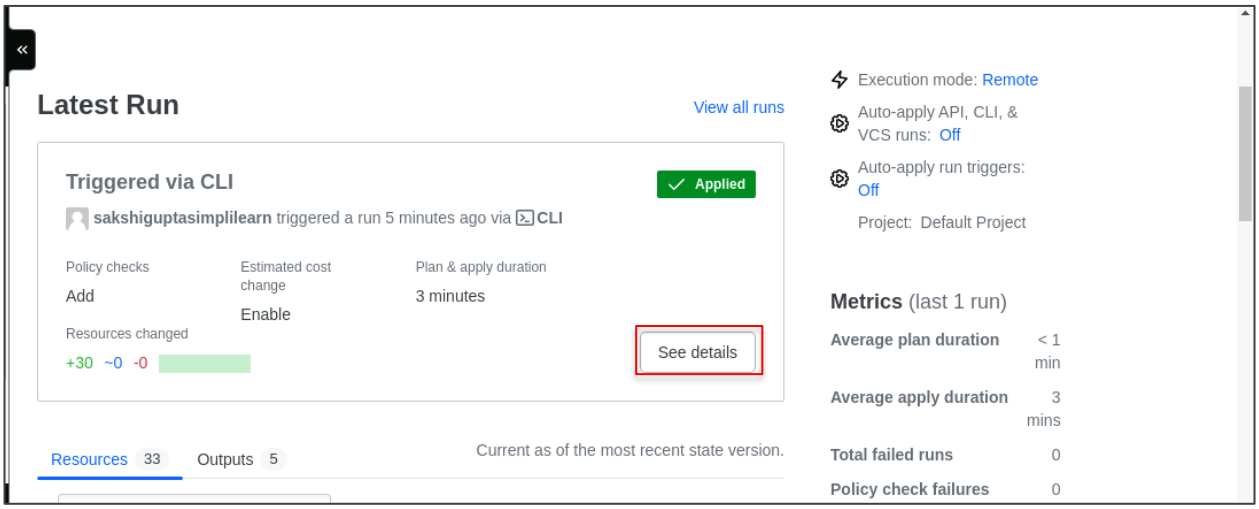


3.5 Click on the workspace **my-aws-app**

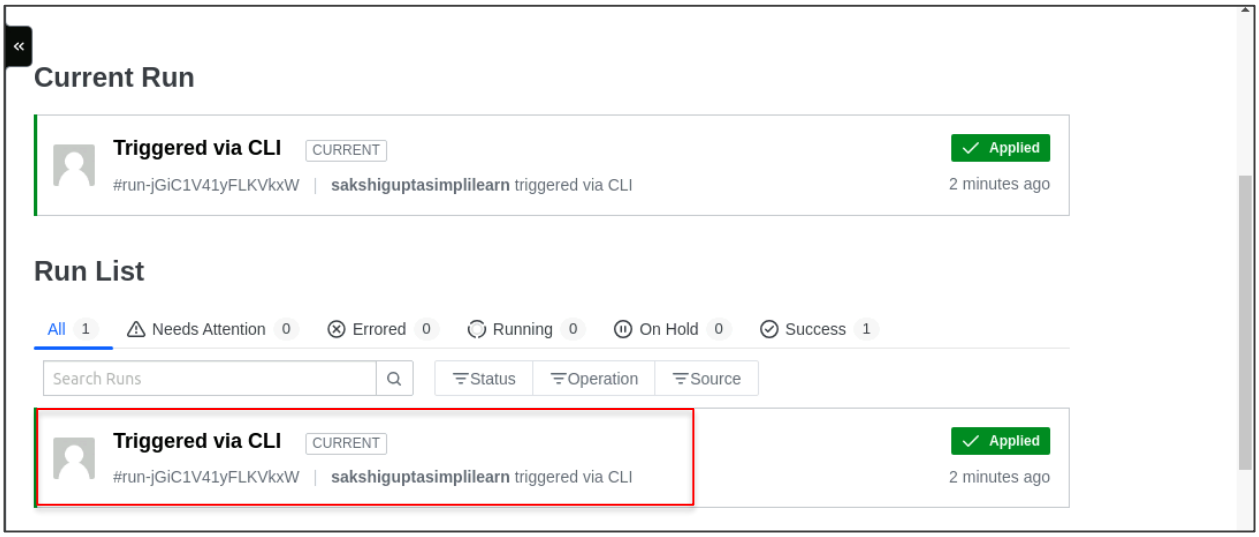




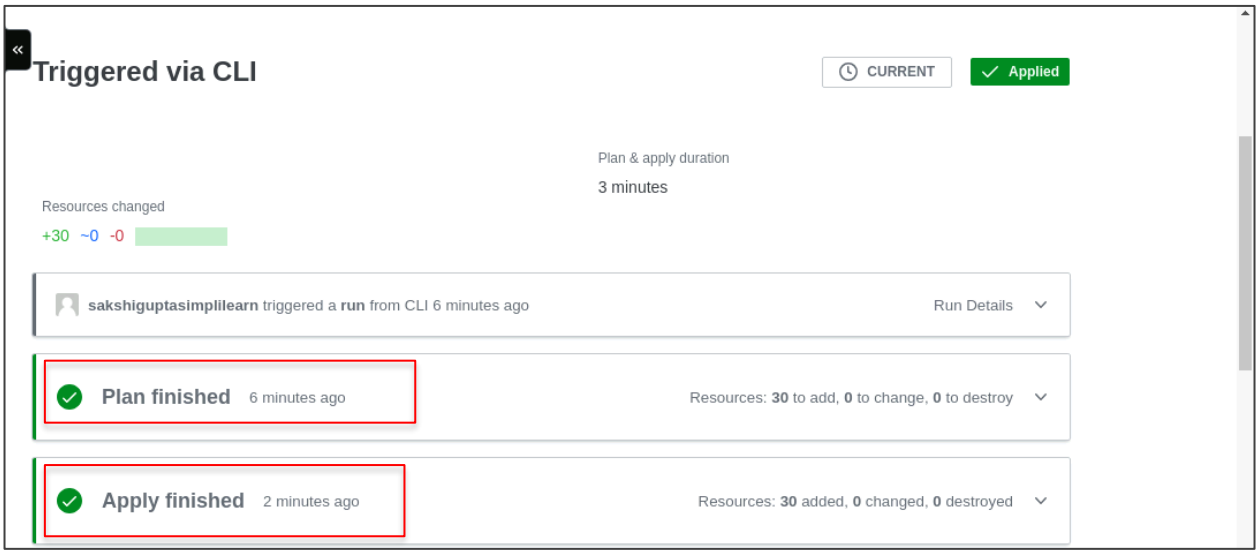
3.6 Scroll down to the **Latest Run** section and click on **See details**

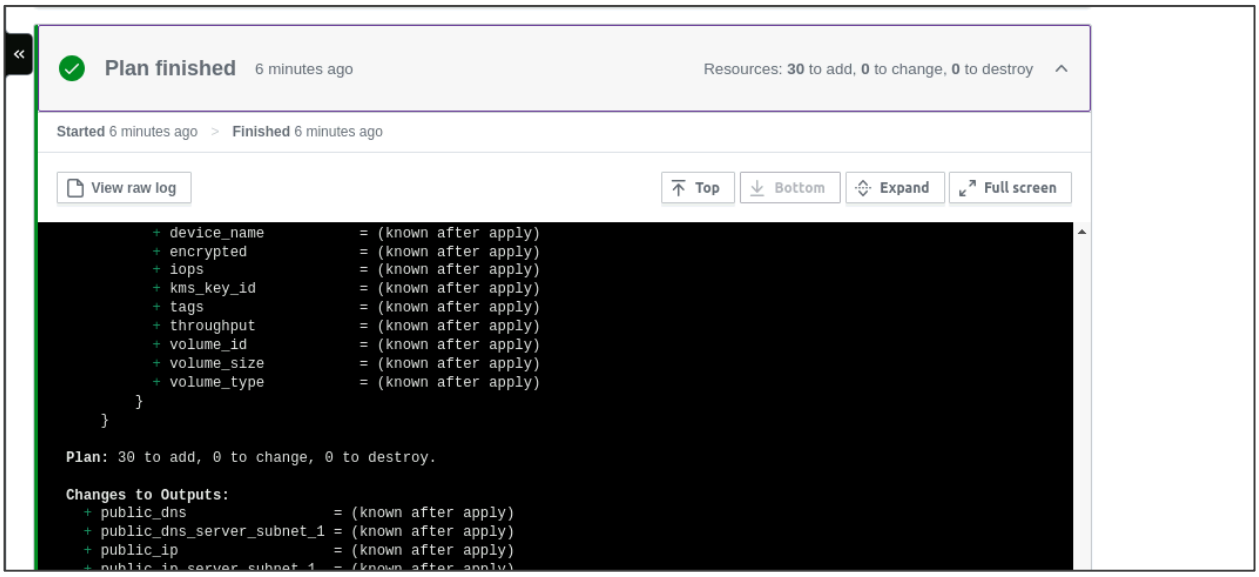


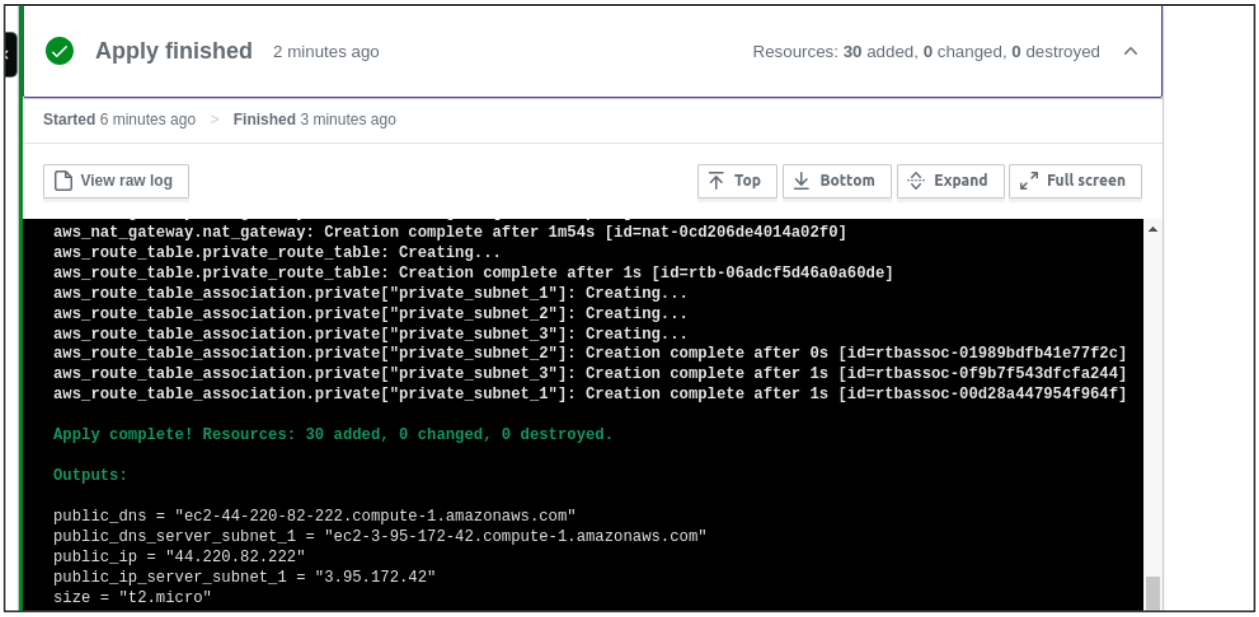
3.7 Click on the latest **Triggered via CLI** under the **Run List**



3.8 Click on **Plan finished** and **Apply finished** to view the command line output on Terraform Cloud via the remote backend

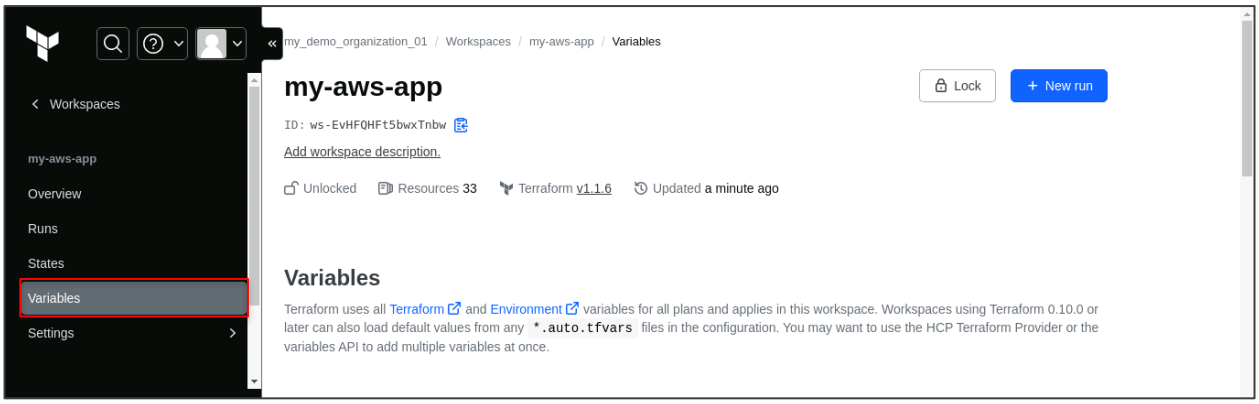

Triggered via CLI
CURRENT
Applied
Plan & apply duration
3 minutes
Resources changed
+30 ~0 -0
sakshiguptasimplilearn triggered a run from CLI 6 minutes ago
Run Details
Plan finished 6 minutes ago
Resources: 30 to add, 0 to change, 0 to destroy
Apply finished 2 minutes ago
Resources: 30 added, 0 changed, 0 destroyed


Plan finished 6 minutes ago
Resources: 30 to add, 0 to change, 0 to destroy
Started 6 minutes ago > Finished 6 minutes ago
View raw log
Top
Bottom
Expand
Full screen
+ device_name = (known after apply)
+ encrypted = (known after apply)
+ iops = (known after apply)
+ kms_key_id = (known after apply)
+ tags = (known after apply)
+ throughput = (known after apply)
+ volume_id = (known after apply)
+ volume_size = (known after apply)
+ volume_type = (known after apply)
}
}
Plan: 30 to add, 0 to change, 0 to destroy.
Changes to Outputs:
+ public_dns = (known after apply)
+ public_dns_server_subnet_1 = (known after apply)
+ public_ip = (known after apply)

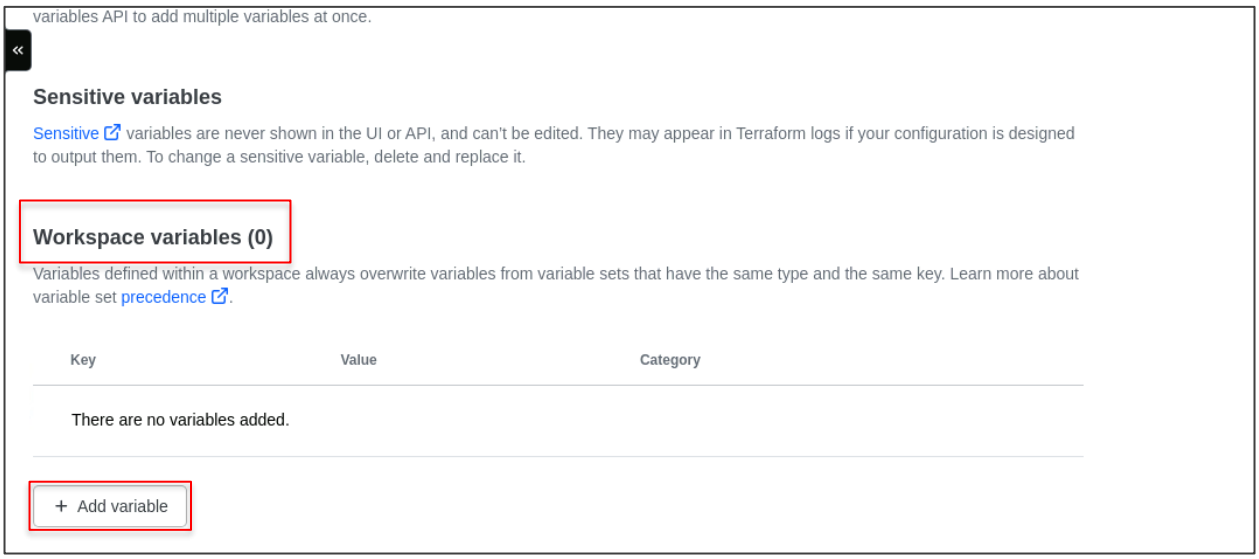


Step 4: Provide secure credentials for remote runs

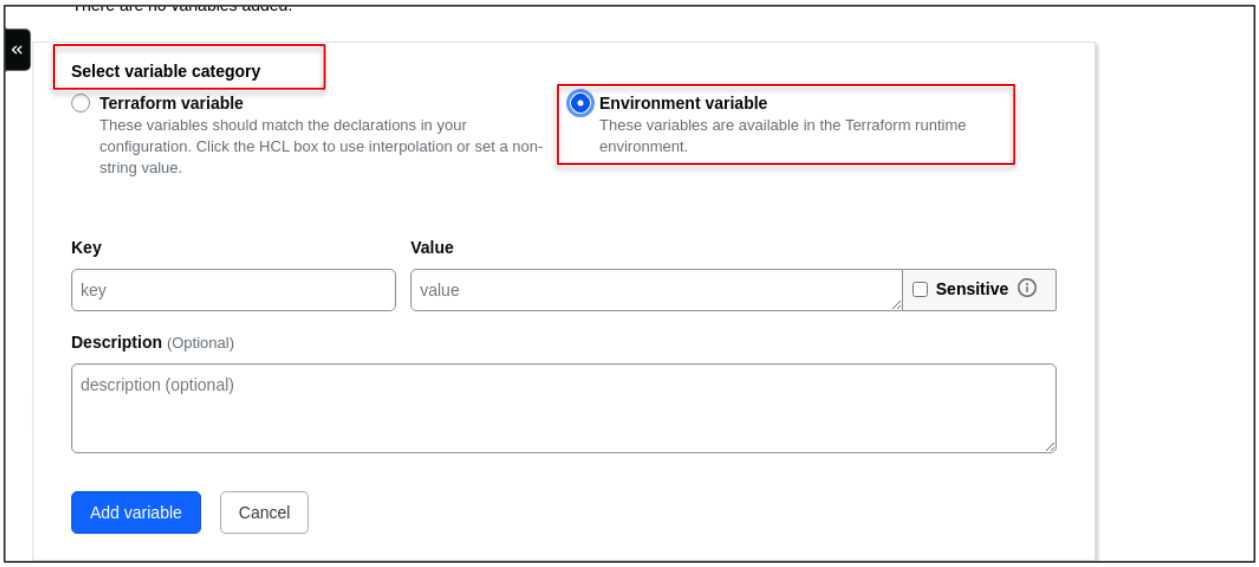
4.1 Go to the **Variables** section of your workspace



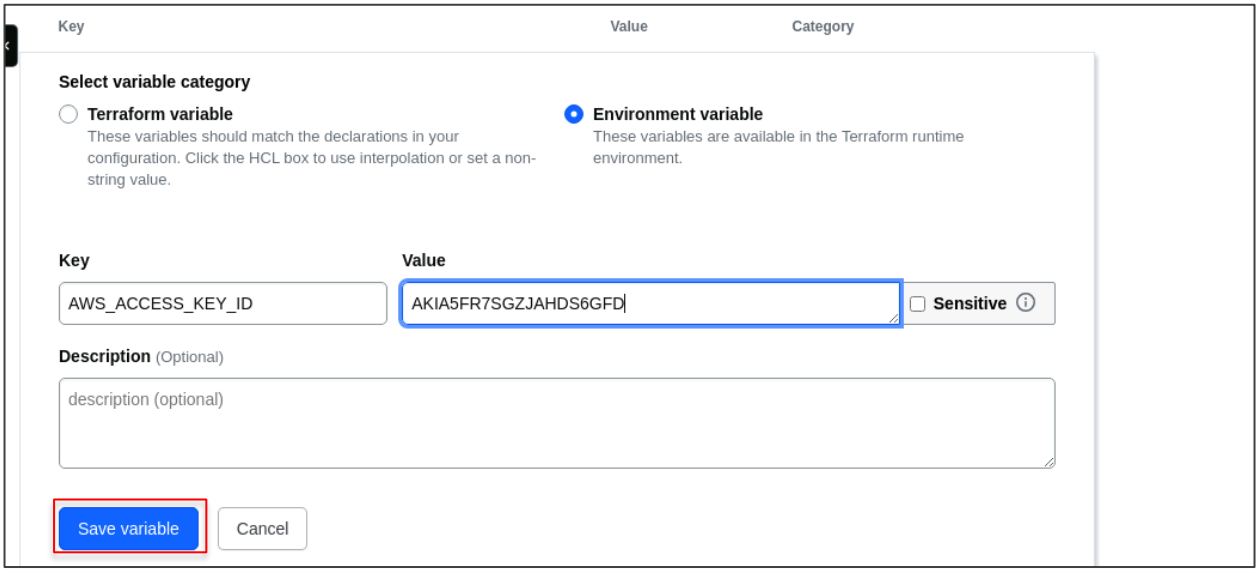
4.2 Scroll down to the **Workspace variables** and click on **Add variable**

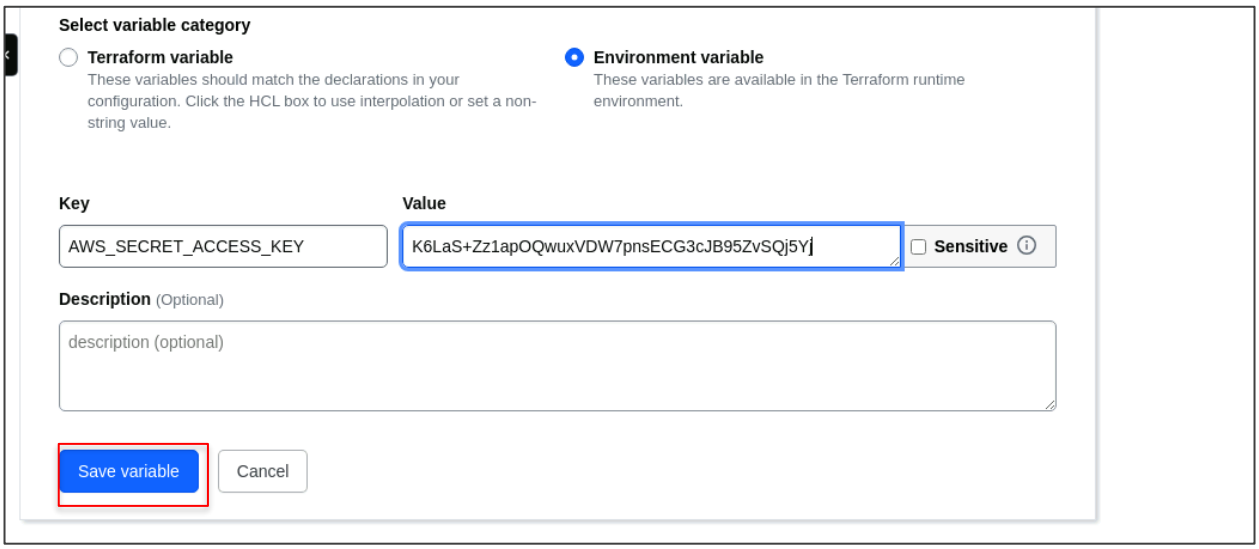


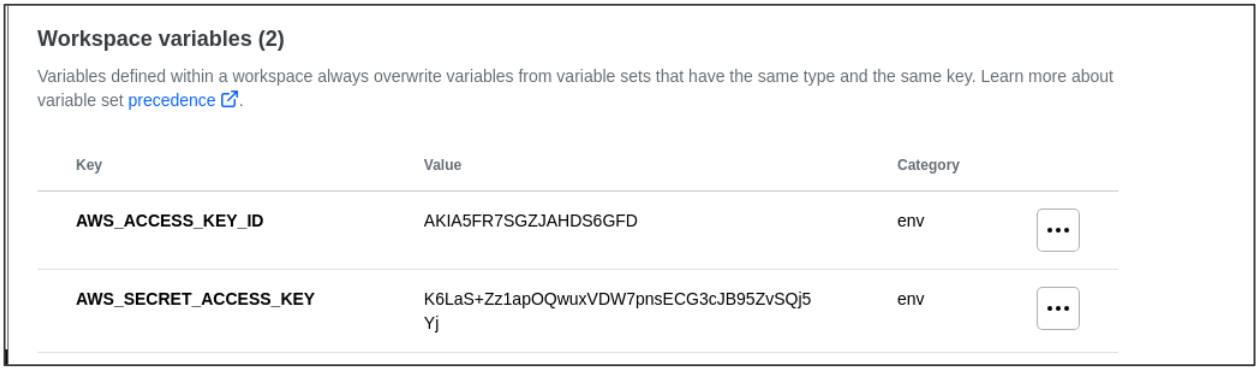
4.3 Select variable category as **Environment variable**



4.4 Add the details for your access key and secret key as shown in the screenshots. After this, click on **Save variable**.







4.5 Go to the terminal in Visual Studio Code and apply the Terraform configuration with the new credentials using the following command:

terraform apply

```
sakshiguptasimp@ip-172-31-22-2:~/Desktop/Terraform$ terraform apply
Running apply in the remote backend. Output will stream here. Pressing Ctrl-C
will cancel the remote apply if it's still pending. If the apply started it
will stop streaming the logs, but will not stop the apply running remotely.

Preparing the remote apply...

To view this run in a browser, visit:
https://app.terraform.io/app/my_demo_organization_01/my-aws-app/runs/run-hvpfGauWSyBDnb7M

Waiting for the plan to start...
```

```
PROBLEMS    OUTPUT    TERMINAL    DEBUG CONSOLE    PORTS

Terraform will perform the following actions:

  # local_file.private_key_pem will be created
  + resource "local_file" "private_key_pem" {
      + content              = (sensitive)
      + directory_permission = "0777"
      + file_permission      = "0777"
      + filename             = "MyAWSKey.pem"
      + id                   = (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.


Do you want to perform these actions in workspace "my-aws-app"?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes
```

```
PROBLEMS    OUTPUT    TERMINAL    DEBUG CONSOLE    PORTS                bash

Do you want to perform these actions in workspace "my-aws-app"?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

local_file.private_key_pem: Creating...
local_file.private_key_pem: Creation complete after 0s [id=34cdb014049a9e0c76a19283afe30e2ef870b717]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

public_dns = "ec2-44-220-82-222.compute-1.amazonaws.com"
public_dns_server_subnet_1 = "ec2-3-95-172-42.compute-1.amazonaws.com"
public_ip = "44.220.82.222"
public_ip_server_subnet_1 = "3.95.172.42"
size = "t2.micro"
sakshiguptasimp@ip-172-31-22-2:~/Desktop/Terraform$
```

Step 5: Remove existing resources with the terraform destroy command

5.1 Clean up all existing resources managed by Terraform using the following command:
terraform destroy

```
sakshiguptasimp@ip-172-31-22-2:~/Desktop/Terraform$ terraform destroy
Running apply in the remote backend. Output will stream here. Pressing Ctrl-C
will cancel the remote apply if it's still pending. If the apply started it
will stop streaming the logs, but will not stop the apply running remotely.

Preparing the remote apply...

To view this run in a browser, visit:
https://app.terraform.io/app/my_demo_organization_01/my-aws-app/runs/run-wii4eSoSKe8w7s8f

Waiting for the plan to start...

Terraform v1.1.6
on linux_amd64
Initializing plugins and modules...
```

5.2 When prompted, approve the destruction by typing **yes**

```
PROBLEMS    OUTPUT    TERMINAL    DEBUG CONSOLE    PORTS

          - volume_size           = 8 -> null
          - volume_type           = "gp2" -> null
        }
    }

Plan: 0 to add, 0 to change, 29 to destroy.

Changes to Outputs:
  - public_dns                 = "ec2-44-220-82-222.compute-1.amazonaws.com" -> null
  - public_dns_server_subnet_1 = "ec2-3-95-172-42.compute-1.amazonaws.com" -> null
  - public_ip                  = "44.220.82.222" -> null
  - public_ip_server_subnet_1  = "3.95.172.42" -> null
  - size                       = "t2.micro" -> null

Do you really want to destroy all resources in workspace "my-aws-app"?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

  Enter a value: yes
```

```
PROBLEMS    OUTPUT    TERMINAL    DEBUG CONSOLE    PORTS

aws_subnet.public_subnets["public_subnet_3"]: Destroying... [id=subnet-031b4ccbab76352ca]
aws_subnet.public_subnets["public_subnet_2"]: Destroying... [id=subnet-0d3714b9ed6230472]
aws_subnet.public_subnets["public_subnet_3"]: Destruction complete after 0s
aws_subnet.public_subnets["public_subnet_1"]: Destruction complete after 0s
aws_subnet.public_subnets["public_subnet_2"]: Destruction complete after 0s
aws_eip.nat_gateway_eip: Destruction complete after 1s
aws_internet_gateway.internet_gateway: Destroying... [id=igw-0935512912e66c367]
aws_internet_gateway.internet_gateway: Destruction complete after 0s
aws_vpc.vpc: Destroying... [id=vpc-01c721fb010bc8ef8]
aws_vpc.vpc: Destruction complete after 1s

Apply complete! Resources: 0 added, 0 changed, 29 destroyed.

sakshiguptasimp@ip-172-31-22-2:~/Desktop/Terraform$
```

By following these steps, you have successfully configured and managed Terraform state using the remote enhanced backend with Terraform Cloud.